TUTORIAL 3 ASSIGNMENT (INFORMATION TECHNOLOGY)

Quiz — Check your understanding

1. **Write a line of Python code** that displays the sum of `468 + 751.`

   print(468 + 751)

   ```
   print(468 + 751)
   1219
   ```

2. **Write a line of Python code** that displays the words `"How are you?"` to the screen.

   print ("How are you?")

   ```
   print("How are you?")
   How are you?
   ```

3. **What is the output** of the following Python statement?

   `print(8 / 3, 4 * 7, 9 + 13, 2 ** 5, 6 * (3 + 2))`

   `O/P:` 2.6666666666666665 28 22 32 30

   ```
   print(8 / 3, 4 * 7, 9 + 13, 2 ** 5, 6 * (3 + 2))
   2.6666666666666665 28 22 32 30
   ```

4. **Write a Python statement** that creates a variable called `size` and assigns the value 77 to it.

   size = 77

   Print(size)

   ```
   size = 77
   print(size)
   77
   ```

5. **What will be the output** of the following Python program?

   ```
   x = 5
    y = 7
    print(abs(x - y) - 10)
    print(int(x ** 2) + 1.4)
    print(round(y + 3.14159, 2))
   ```

   ```
   x = 5
   y = 7
   print(abs(x - y) - 10)
   -8
   print(int(x ** 2) + 1.4)
   26.4
   print(round(y + 3.14159, 2))
   10.14
   ```

6. **What is the output** of the following Python program?

```
a = 31
 b = 7
 print(a // b, a % b)
o/p: 4 3
```

```
a = 31
b = 7
print(a // b, a % b)
4 3
```

7. **Write a Python program** to convert `250` minutes into hours and minutes and print both values.

minutes = 250

hours = minutes // 60

remaining_minutes = minutes % 60

print("Hours:", hours)

print("Hours:", hours)print("Minutes:", remaining_minutes)

```
minutes = 250
hours = minutes // 60
remaining_minutes = minutes % 60
print("Hours:", hours)
Hours: 4
print("Minutes:", remaining_minutes)
Minutes: 10
```

8. **What is the output** of the following Python program?

```
str1 = "it is what it is"
print(str1[-9:])
```

```
str1 = "it is what it is"
print(str1[-9:])
hat it is
```

9. Additional Exercises

assume `a = 1` and `b = 1.5`. For each condition, state whether it evaluates *True* or *False*.

- `3 * a == 2 * b – TRUE`

- `(a < b) or (b < a) - TRUE`
- `not (a < b) or not (a < (b - a)) - TRUE`
- `((a == b) or not (b < a)) and ((a < b) or (b == a + 1)) - TRUE`

10. determine whether each condition is *True* or *False*.

- `'Ab' == 'ab' - FALSE`
- `'Harry' < 'harry' - TRUE`
- `not (('B' == 'b') or ("Big" < "big")) - FALSE`
- `isinstance(32, float) - FALSE`
- `isinstance(32, int) - TRUE`
- `"yt" in "Python" - TRUE`
- `"knight".startswith('n') - FALSE`
- `True or False - TRUE`
- `True and False - FALSE`
- `not True - FALSE`

11. **Decision Structures . Determine the output displayed.**

```
a = 2
b = 3
c = 7
if(a*b)<c:
   b = a
else:
   c = a+b+c
print(a,b,c)
```

O/P: 2 2 7

12. Assume the response is B

```
letter = input("Enter A, B, or C: ")
letter = letter.upper()
if letter == "A":
   print("A, my name is Alice.")
elif letter == "B":
   print("To be, or not to be.")
```

```
elif letter == "C":
    print("Oh, say, can you see.")
else:
    print("You did not enter a valid letter.")
```

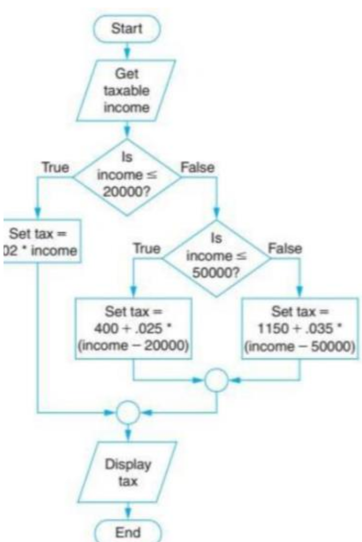
```
================ RESTART: C:/Users/an
Enter A, B, or C: A
A, my name is Alice.

================ RESTART: C:/Users/an
Enter A, B, or C: B
To be, or not to be.

================ RESTART: C:/Users/an
Enter A, B, or C: C
Oh, say, can you see.

================ RESTART: C:/Users/an
Enter A, B, or C: V
You did not enter a valid letter.
|
```

13.
```
isvowel = False
letter = input("Enter a letter: ")
letter = letter.upper()

      if letter in "AEIOU":
     isvowel = True
     if isvowel:
     print(letter, "is a vowel.")
elif not (65 <= ord(letter) <= 90):
     print("You did not enter a letter")
else:
     print(letter, "is a consonant.")
```

```
================ RESTART: C:
Enter a letter: G
G is a consonant.

================ RESTART: C:
Enter a letter: A
A is a vowel.

================ RESTART: C:
Enter a letter: I
I is a vowel.

================ RESTART: C:
Enter a letter: E
E is a vowel.

================ RESTART: C:
Enter a letter: O
O is a vowel.

================ RESTART: C:
Enter a letter: U
U is a vowel.

================ RESTART: C:
Enter a letter: Z
Z is a consonant.
```

14. **Income Tax (flowchart)**

Using the flowchart provided, write a Python program that reads a person's *taxable income*, computes the state income tax according to the diagram, and then displays the tax. (Do not include explanations here—just your program.)



```
income = float(input("Enter taxable income: "))
if income <= 20000:
 tax = 0.02 * income
elif income <= 50000:
 tax = 400 + 0.025 * (income - 20000)
else:
 tax = 1150 + 0.035 * (income - 50000)
print("Tax:", tax)
```

```
================ RESTART: C:/Use
Enter taxable income: 20000
Tax: 400.0

================ RESTART: C:/Use
Enter taxable income: 10000
Tax: 200.0

================ RESTART: C:/Use
Enter taxable income: 100000
Tax: 2900.0

================ RESTART: C:/Use
Enter taxable income: 50000
Tax: 1150.0
```